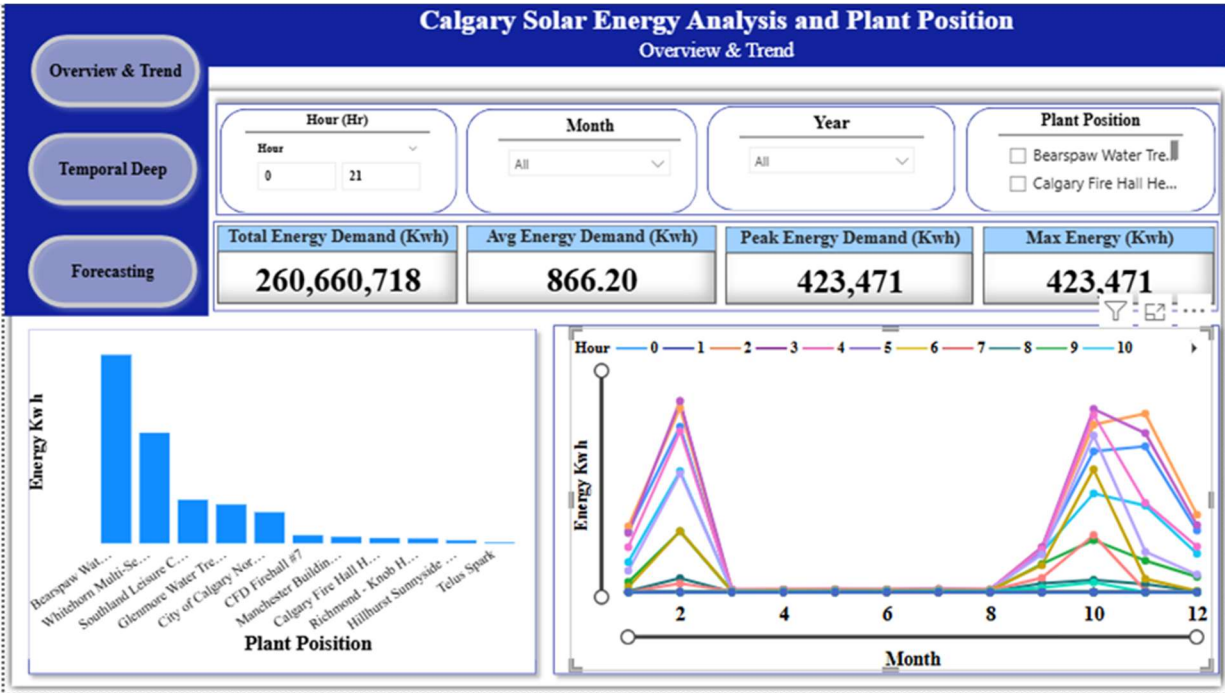




CALGARY SOLAR ENERGY ANALYTICS



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1. Introduction

1.1 Overview

Calgary is rapidly expanding its renewable energy infrastructure, with solar PV installations playing an increasingly important role in the city's energy mix. While solar energy is environmentally sustainable, it is naturally seasonal and volatile, which poses challenges for accurate forecasting, operational planning, and grid management. Reliable predictions are crucial for:

- Ensuring operational efficiency in energy management
- Facilitating seamless integration of renewable energy into the grid
- Supporting Calgary's Net-Zero 2050 goals

1.2 Aim / Objective

The main objective of this project is to convert historical solar PV energy data into actionable insights for data-driven decision-making. Specifically, this project seeks to:

- Analyze historical solar energy trends across Calgary PV installations
- Identify seasonal, daily, and hourly patterns in energy generation
- Forecast short- and medium-term energy production at an hourly resolution
- Provide recommendations to optimize operational planning, load management, and renewable energy integration

1.3 Scope of Work

To achieve these objectives, the project covers the following key areas:

- Data Understanding & Cleaning: Assess data quality, handle missing values and outliers, and prepare data for analysis
- Exploratory Data Analysis (EDA): Examine temporal patterns, seasonal trends, volatility, and site-specific behaviors
- Feature Engineering: Create lag features, rolling averages, sunrise/sunset indicators, and site-level metadata
- Forecasting & Modeling: Develop predictive models using classical approaches (Simple Exponential, ARIMA, SARIMA, PROPHET)
- SQL & ETL Pipelines: Automate extraction, transformation, and loading of large-scale datasets using SQLAlchemy for reproducibility and scalability
- Business Intelligence & Visualization: SQL and PowerBi connected , Communicate insights through dashboards for operational and strategic decision-making

1.4 Problem Statement / Challenges

Despite the dataset's richness, several challenges exist:

- **Seasonal Imbalance:** Winter months exhibit lower and inconsistent energy output, while summer months show high and stable production
- **Outliers:** Extreme low or high values occur during unusually sunny or cold days
- **Zero-output Periods:** Nighttime, maintenance, or sensor faults create gaps in the data
- **Missing Feature Values:** Rolling and lag features have missing values at the start of their windows
- **Site-specific Data Gaps:** Not all installations have complete historical records, leading to inconsistencies

1.5 Solution / Approach

To overcome these challenges, an end-to-end analytics workflow was implemented:

1. **Data Extraction & Cleaning:** Standardized and prepared data using SQLAlchemy pipelines
2. **Exploratory Data Analysis:** Identified temporal trends, seasonal patterns, volatility, and outliers
3. **Feature Engineering:** Generated rolling averages, lag features, and temporal flags to improve model performance
4. **Forecasting & Modeling:** Applied Simple Exponential, ARIMA, SARIMA, and Prophet models for accurate predictions
5. **Visualization & Reporting:** Built dashboards for operational insights, peak load scheduling, and long-term planning

Key Insight:

Despite seasonal volatility and occasional anomalies, Calgary’s solar PV dataset contains consistent temporal patterns that enable reliable forecasting and support renewable energy integration.

2. Data Understanding & Challenges

2.1 Dataset Overview

The dataset captures hourly solar PV energy generation across multiple installations in Calgary from 2015 to 2024, containing approximately 258,000 rows. Each row represents an hourly observation for a specific PV installation.

2.2 Key Columns / Features

Column	Description	Purpose
datetime	Timestamp of energy measurement	For temporal analysis (hourly, daily, seasonal)
name	PV installation name	Identifies the site for site-level analysis

address	Location of installation	Enables geographic insights and clustering
energy_kwh	Hourly energy output in kWh	Target variable for forecasting
capacity	PV system capacity (kW)	Useful for normalization and comparisons
installation_date	Date when the installation was commissioned	Helps account for site age and seasonal patterns

Engineered / Derived Features:

- Lag Features: Previous energy outputs (e.g., 24h, 168h) to capture daily and weekly seasonality
- Rolling Statistics: Moving averages (3h, 6h, 24h) to smooth short-term fluctuations
- Sunrise/Sunset Flags: Indicator for daylight vs nighttime
- Site Metadata: Capacity, location, and installation date for feature enrichment

2.3 Dataset Size & Structure

- Rows: ~258,000 hourly observations across multiple sites
- Columns: 6–10 core columns + engineered features
- Granularity: Hourly, enabling fine-resolution forecasting and temporal analysis
- Time Horizon: 2015–2024 (~9–10 years), allowing robust trend and seasonality modeling

Example Row:

datetime	name	address	energy_kwh	capacity	installation_date	dashboard_url
2023-06-15 12:00	Bearspaw Plant	Calgary NW	1,245 kWh	500 kW	2015-08-10	link-to-dashboard

2.4 Data Challenges

1. Seasonal Imbalance: Winter months show low and fluctuating output; summer months peak consistently.
2. Outliers: Extreme low/high values due to sunny or cold days, which may skew analysis.
3. Zero-output Periods: Nighttime, maintenance, or sensor errors create gaps.
4. Missing Values in Features: Rolling and lag features have missing values at the start.
5. Site Data Inconsistency: Some installations have incomplete historical records, affecting temporal continuity.

2.5 Key Insights

- Hourly granularity allows precise short-term forecasting
- Nearly a decade of data supports long-term trend analysis

- Site-specific behaviors, daily cycles, and seasonal patterns are crucial for reliable predictions and operational planning
- Despite challenges, the dataset provides a rich foundation for feature engineering and predictive modeling

3. Exploratory Data Analysis (EDA)

Solar energy generation in Calgary follows predictable temporal patterns, but winter months introduce risk due to high volatility. Understanding these trends supports reliable forecasting, operational planning, and resource scheduling.

Temporal Analysis:

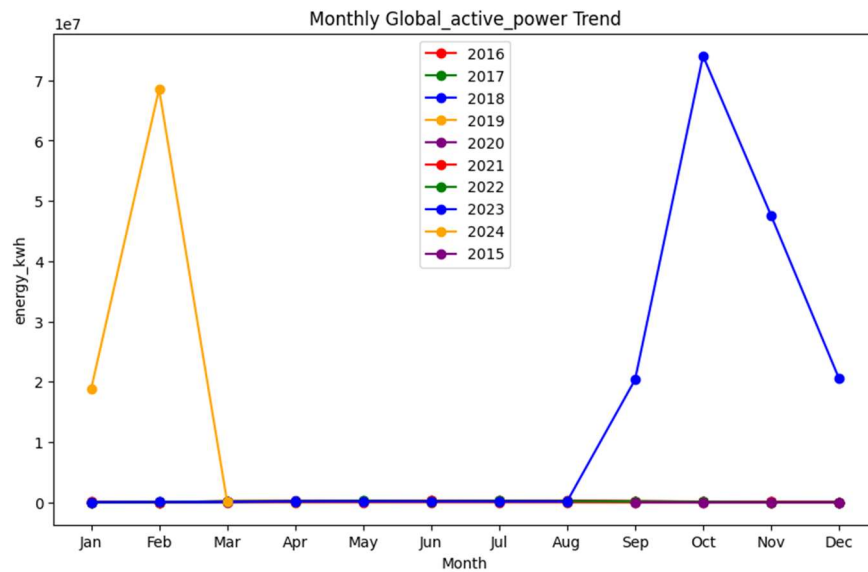
- Hourly energy curves show daytime dominance with peaks near midday
- Monthly aggregations reveal seasonal cycles, with February showing the highest average (~410 kWh) and January the lowest (~44 kWh)
- Summer vs winter comparison shows winter has more variability and extreme outliers

Distribution & Volatility:

- Histogram analysis: most days have moderate energy, extreme high values are rare
- Rolling statistics (3h, 24h) highlight short-term fluctuations vs long-term trends
- Volatility analysis (100-day rolling window) shows:
 - Early period: low, stable energy
 - Mid-period: increasing fluctuations
 - Late period: highest volatility

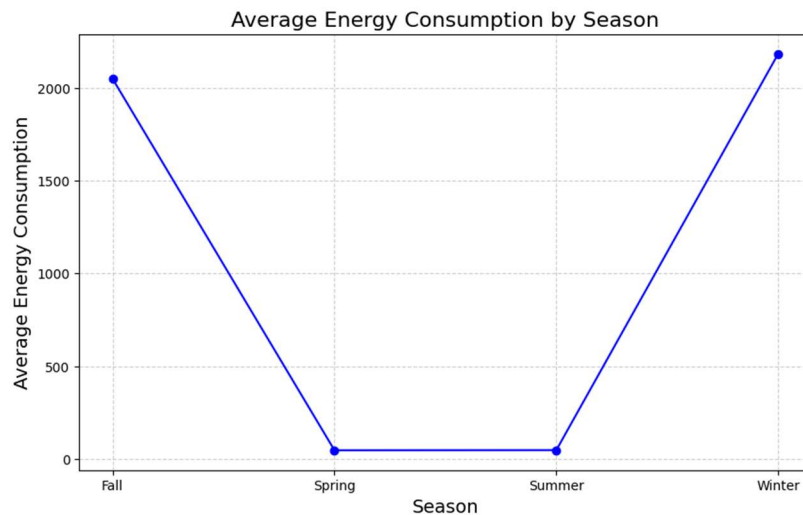
3.1.Evaluation of data by month, year, and calculate the energy Kwh

From the data, the highest average power generation occurs in February, at approximately 410 kWh, while the lowest average power generation is observed in January, at around 44 kWh. This visualization reveals a clear seasonal pattern in energy generation. Output peaks in February, followed by strong performance in October and November, while January consistently shows the lowest production. Understanding these month-to-month and year-over-year trends supports better energy forecasting, scheduling, and operational planning.



3.2.Average Energy Consumption by seasonal

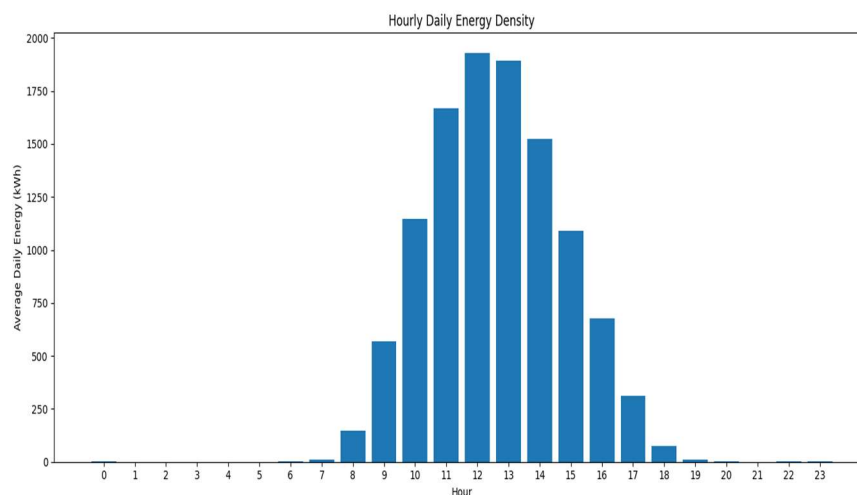
This visualization highlights periods of peak and low energy generation, supporting effective seasonal energy planning and operational decision-making. Energy consumption remains relatively high during winter and continues consistently through spring and summer, indicating stable demand across multiple seasons rather than sharp seasonal drops.



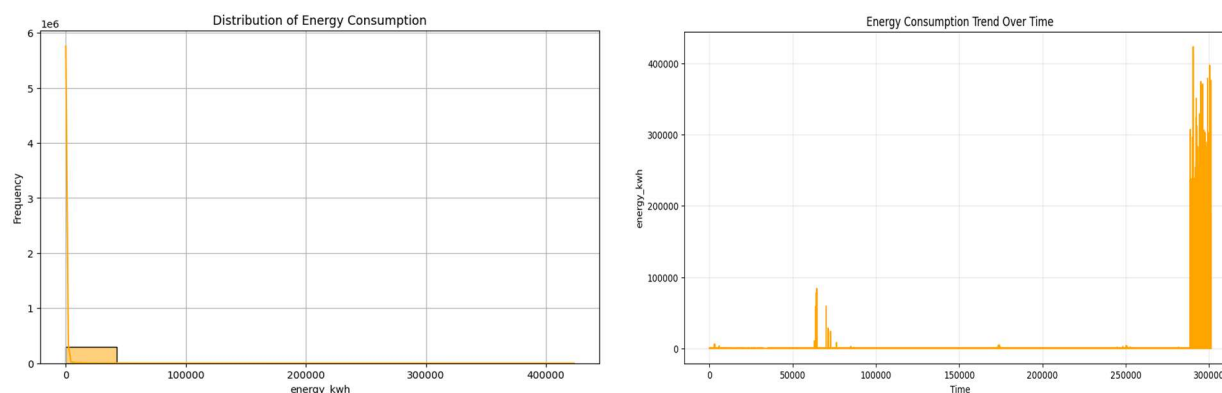
Energy generation demonstrates sustained performance across seasons, with strong output in winter and continued consistency through spring and summer. This stability reduces seasonal risk and enables more reliable energy forecasting, capacity planning, and operational scheduling.

3.3.Hourly Daily Energy Density

This chart shows how average daily energy (kWh) varies by hour of the day. Energy levels are lowest during the early morning (around 4–5 AM), increase steadily after 6 AM, and peak around midday to early afternoon. In some cases, a smaller evening peak appears as residential demand rises. These patterns highlight predictable daily energy cycles that are useful for demand forecasting, load balancing, and operational planning.



3.4. Distribution of energy consumption & over time

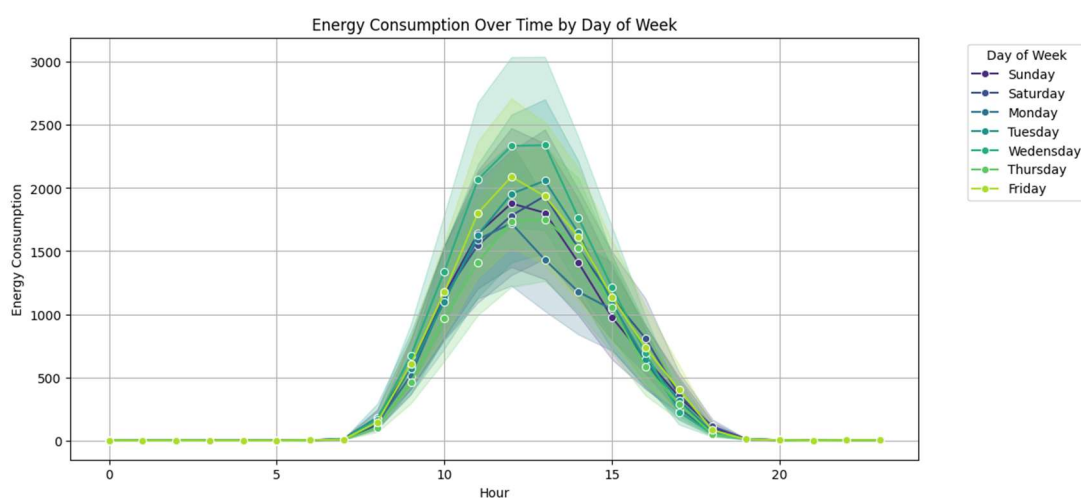


This histogram shows how frequently different levels of energy consumption occur. Most values are concentrated at the lower end, indicating that typical days or units use moderate energy, while very high consumption (>200,000 kWh) is rare. The few extreme values highlight outliers or unusually high usage. Understanding this distribution helps identify normal versus abnormal energy use and supports targeting efficiency improvements or monitoring high-consumption periods.

This line plot depicts how energy consumption evolves over time. Overall, there is an upward trend, with fluctuations representing seasonal variations, events, or anomalies. Peaks and dips suggest periods of higher or lower demand, while the general slope indicates growing energy usage. Monitoring these trends is valuable for forecasting future demand, detecting seasonal patterns, and assessing the effectiveness of energy-saving initiatives.

3.5. Energy Consumption Over Time by Day of Week

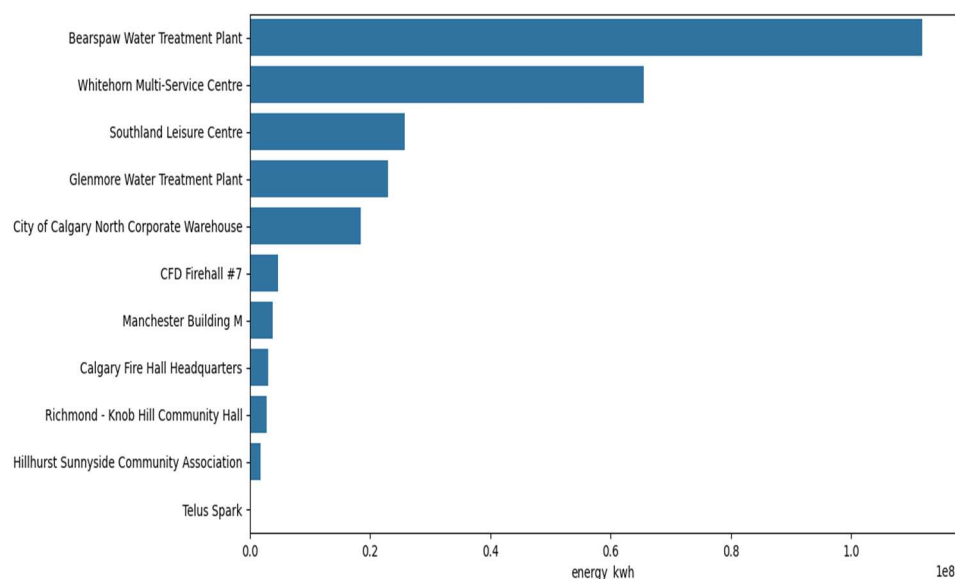
This multi-line time series shows daily energy consumption by day of the week. Weekly patterns are evident, with consumption higher on weekdays (Monday–Friday) and lower on weekends (Saturday & Sunday). Peaks typically occur on Mondays and Tuesdays, while Fridays show a slight decline. Over the observed period, the weekly seasonality dominates any longer-term trend. Understanding these day-of-week patterns supports load planning, energy forecasting, and targeted energy-saving strategies, such as reducing base load during weekends.



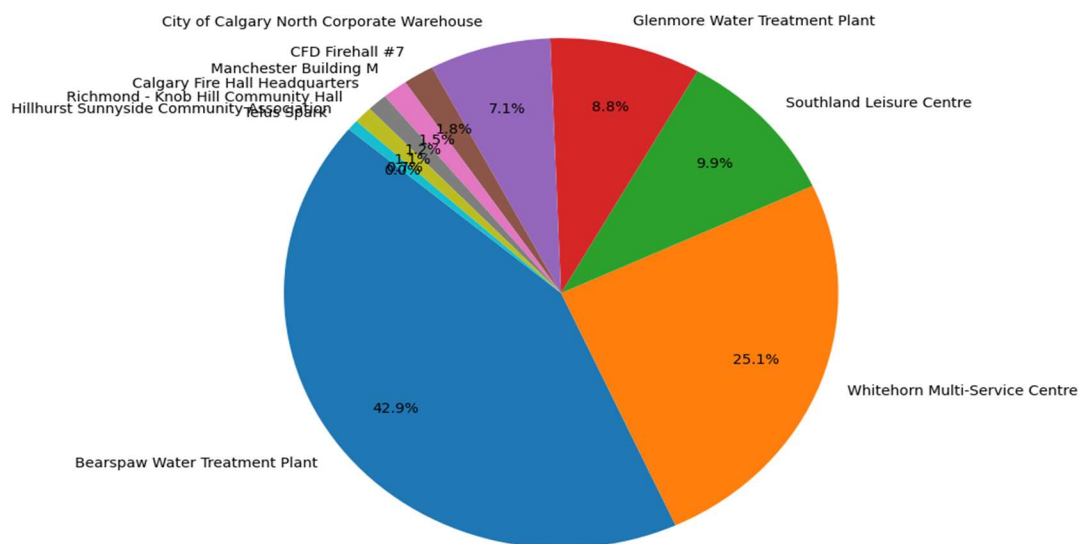
3.6. Energy Production by Facility in Alberta

The Bearspaw Water Treatment Plant is the largest contributor at 42.9%, followed by Glenmore Water Treatment Plant at 25.1%, together accounting for 68% of total energy production. The Southland Leisure Centre contributes nearly 10%, while community centers and municipal buildings, such as Hillhurst Sunnyside and various fire halls, make up the remaining smaller percentages. Energy production is dominated by major water treatment plants, highlighting their significant generation capacity. Secondary contributors, including leisure centers and municipal facilities, provide additional energy support. This profile likely reflects municipal renewable or cogeneration systems across City of Calgary facilities.

Total Solar Power Production per Plant



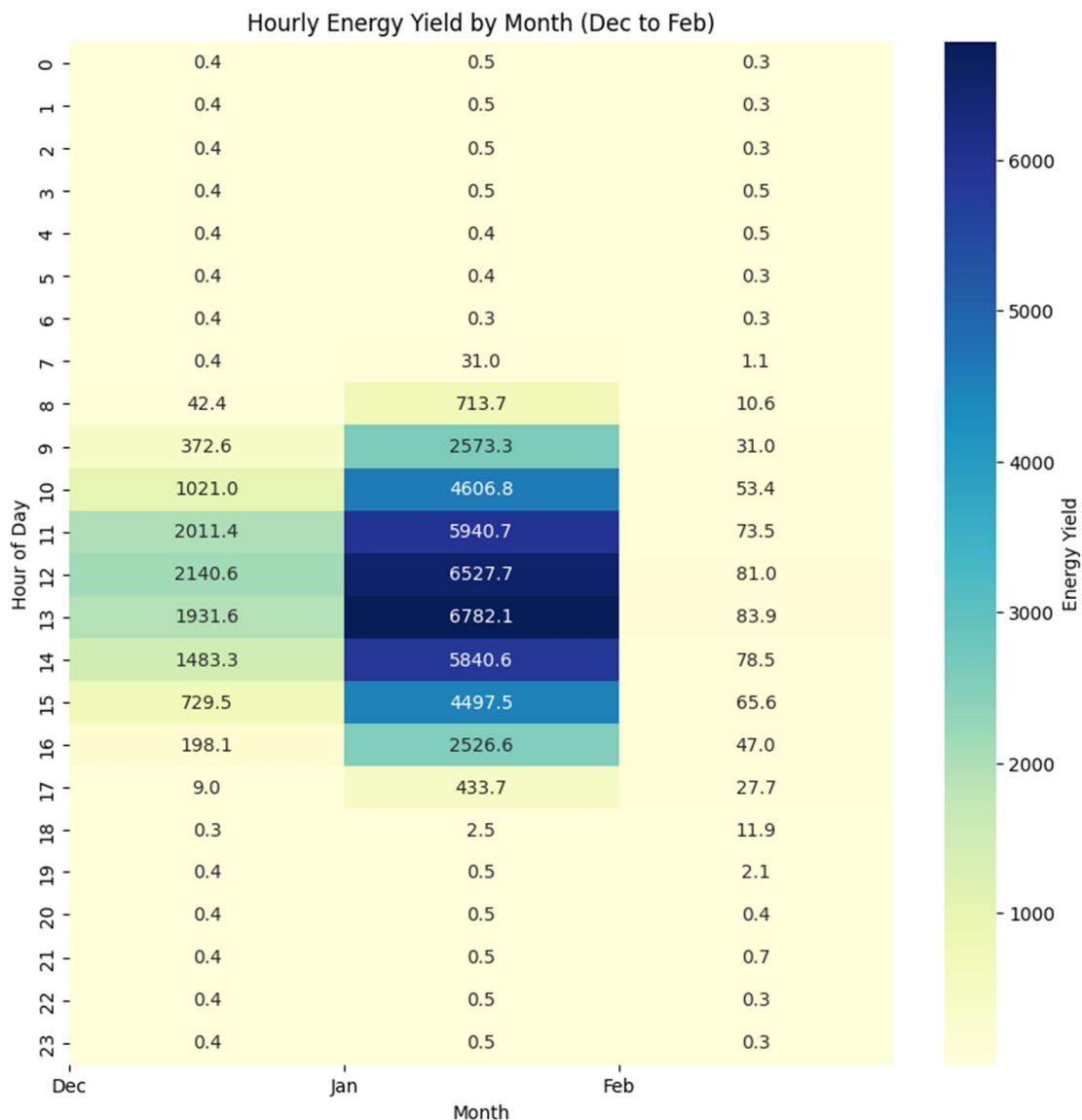
Total Energy Production Distribution by Plant



3.7.Hourly Energy Yield by Month (Dec–Feb)

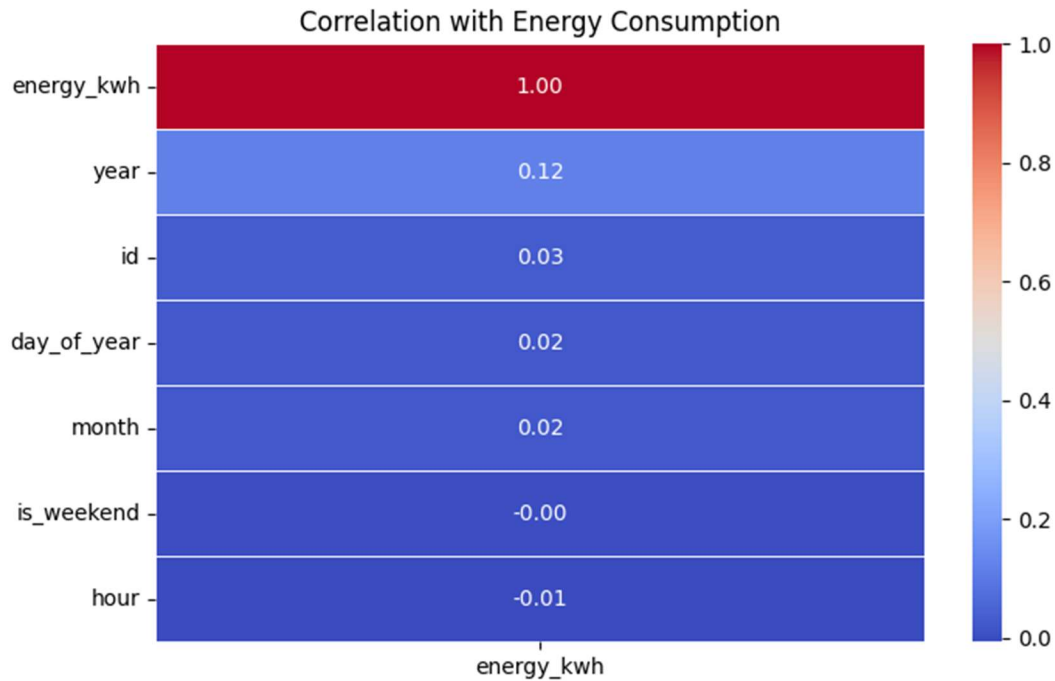
The data shows hourly energy production for the winter months of December, January, and February. Production is very low overnight (~0.4 kWh), rises sharply in the morning, peaks around midday (12–14 hours), and drops again in the late afternoon. This pattern reflects the classic solar energy generation profile. Monthly differences are apparent, one month reaches peaks of ~6,782 kWh, while others peak around ~2,140 kWh, likely due to variations in sunlight, cloud cover, or

solar angles. Winter solar production is limited to roughly 8 AM–4 PM, highlighting the importance of daylight hours for energy planning and forecasting.



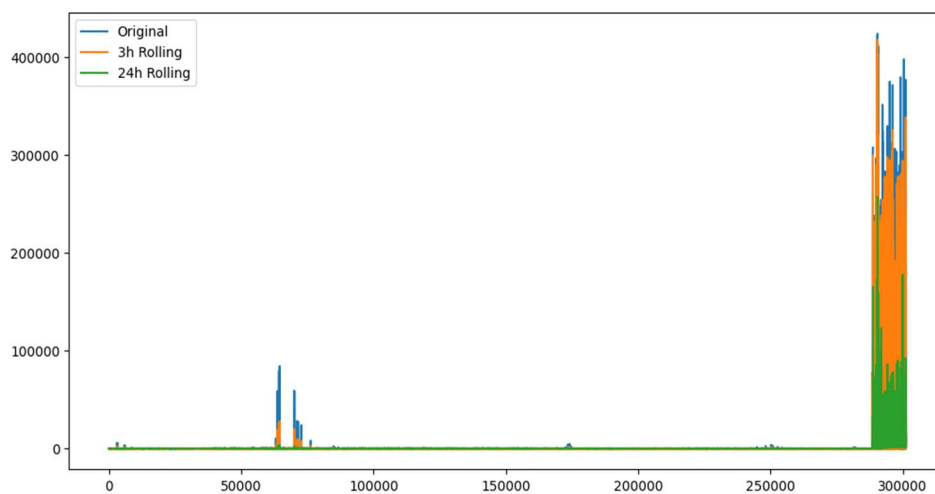
3.8. Correlation Analysis of Energy Consumption

The analysis shows weak overall relationships between energy consumption and the examined time-based variables. The strongest correlation is with year (0.12), suggesting a slight upward trend over time. Other variables, including hour (-0.01), month (~0), and is_weekend (~0), show almost no linear correlation. Energy consumption patterns are complex and largely non-linear. Time-based factors alone do not strongly predict usage, implying that other variables such as temperature, occupancy, or equipment operation are likely more influential. Simple linear models based solely on these variables would perform poorly, highlighting the need for more sophisticated modeling approaches.



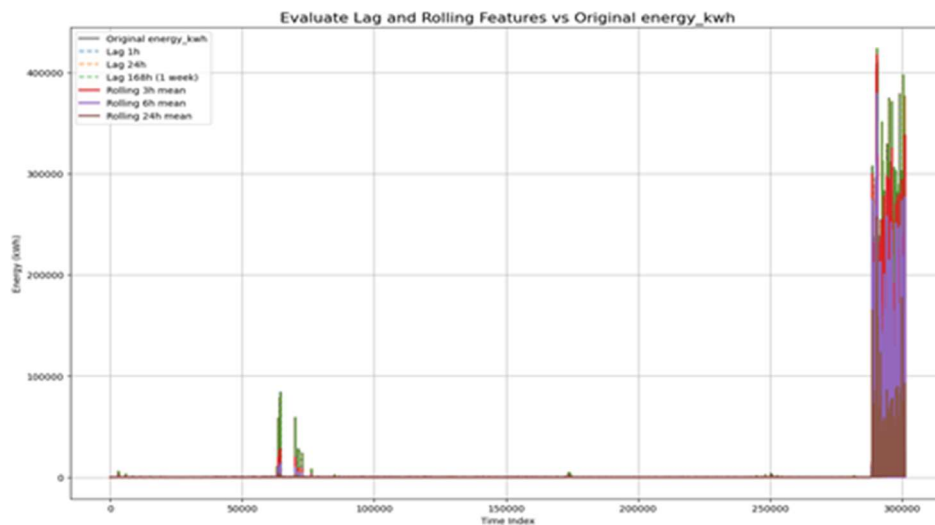
3.9.Original, 3h Rolling, 24h Rolling

This plot compares energy consumption with different smoothing techniques. The original data shows high short-term volatility. The 3-hour rolling average smooths short-term fluctuations while preserving daily patterns, whereas the 24-hour rolling average removes daily cycles entirely, revealing the long-term trend. Rolling averages help separate noise from the underlying signal. Shorter windows capture intra-day variations, while longer windows highlight overall trends, aiding long-term energy planning and pattern recognition.



3.10. Evaluate Lag and Rolling Features vs Original energy_kwh

This plot compares the original energy consumption with lagged and rolling features. Lag 24h captures daily seasonality, lag 168h captures weekly patterns, and rolling averages (3h, 6h, 24h) progressively smooth the data. Understanding which lags and rolling windows align with patterns helps in feature engineering for predictive models, improving forecasting accuracy, anomaly detection, and capacity planning.



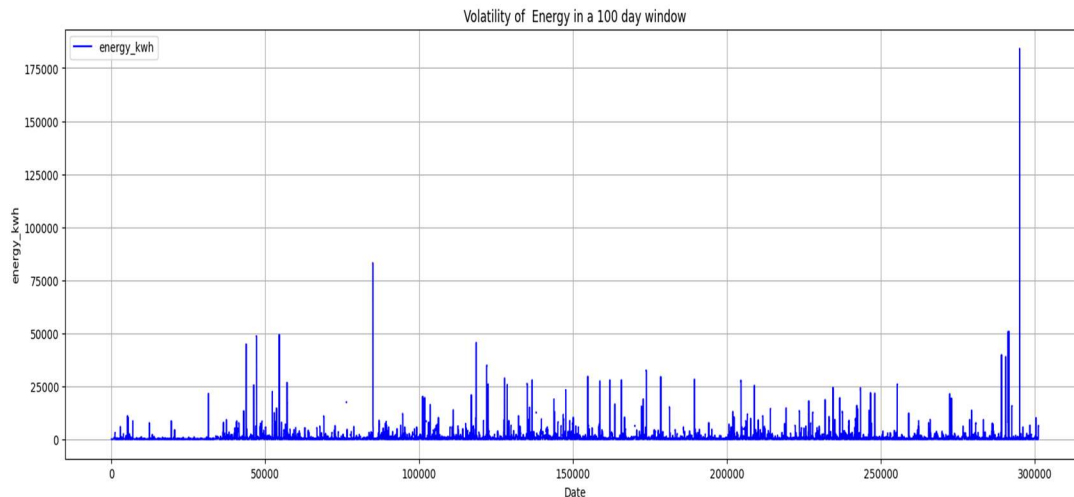
3.11. Volatility of Energy in a 100-day Window

This time series visualizes energy consumption volatility using a 100-day rolling window. Volatility measures fluctuations in energy use, with peaks indicating periods of high variation and troughs showing stable consumption.

Key Insights:

- Early period shows low, stable volatility.
- Mid-period exhibits rising volatility with noticeable spikes.
- Late period reaches the highest volatility, indicating the most unstable energy use.

Understanding volatility is critical for risk assessment, operational planning, and forecasting. High volatility periods require careful grid management and demand response strategies, while stable periods allow for routine maintenance and predictable planning. Cyclical spikes may reflect seasonal effects, weather variability, or operational changes.



Both plots are essential for time series analysis, allowing decomposition of noise, recognition of daily/weekly cycles, and preparation for forecasting models. They guide decisions on which features and smoothing methods to include for better predictions and operational planning.

4. SQL Analytics Layer

4.1 Objective

The goal of this SQL analysis was to extract actionable insights from 10 years of Calgary solar PV energy data. Key questions included:

- Which PV sites perform the best and which need improvement?
- What are the peak energy production hours during the day?
- How do weekdays compare to weekends in energy output?
- Which months or seasons show high variability?
- Are there unusual high-energy days that require attention?
- How does energy production evolve over longer periods (daily, monthly, yearly)?

Purpose: Enable reproducible, database-driven analysis to support operational planning, scheduling, and renewable integration decisions.

4.2 Approach / Actions

To answer these questions, the following SQL-driven workflow was applied:

1. Data Preparation: Filtered the dataset (`calgary_energy_preprocessed`) to ensure clean, consistent data. Removed invalid or missing values where necessary and indexed by datetime.
2. Total & Average Energy per Site: Aggregated energy production by PV installation to identify high- and low-performing sites.
3. Hourly Energy Analysis: Calculated average energy per hour across all sites to detect peak production periods.
4. Weekday vs Weekend Comparison: Grouped data by weekday/weekend to identify behavioral patterns in energy generation.
5. Records Above Daily Average: Compared each record against the daily average to detect unusual high-energy events.
6. Monthly & Seasonal Aggregation: Calculated monthly and seasonal totals and averages to observe long-term trends and seasonal volatility.
7. Rolling & Lag Features: Generated 3-hour rolling sums, lag differences, and site-specific peak hours to capture short-term fluctuations and operational load patterns.
8. Daily, Weekly, Yearly Analysis: Aggregated energy by day of week, day of year, month, and year to support strategic planning.

4.3 Key Results & Insights

Observation	What I Learned	Business Impact / Recommendation
Total energy per site	Site A consistently produces the most energy, while Site B lags	High-performing sites indicate successful operations; underperforming sites require review
Peak hours	Most energy is generated between 2 PM – 6 PM	Resource allocation and operational focus should target these critical hours
Weekday vs weekend	Weekdays generally have higher energy; some sites spike on weekends	Adjust weekend operations or investigate anomalies to optimize output
Records above daily average	Certain days exceed daily energy averages	These days highlight high-demand periods, informing capacity planning and anomaly detection
Monthly / seasonal trends	Winter and summer months exhibit higher variability	Seasonal planning can optimize performance and reduce energy waste
Top 5 peak hours per site	Each site has unique peak hours	Tailored scheduling at the site level improves production efficiency

4.4 SQL Queries

Example 1: Total & Average Energy per Site

```
SELECT name, SUM(energy_kwh) AS total_energy, AVG(energy_kwh) AS avg_energy
FROM first_10_years
GROUP BY name;
```

Over the first decade, Site A led in energy production, while other sites lagged behind, highlighting areas for operational improvement.

Example 2: Peak Hourly Energy

```
SELECT hour, AVG(energy_kwh) AS avg_hourly_energy
FROM first_10_years
GROUP BY hour
ORDER BY avg_hourly_energy DESC;
```

Peak energy occurs between 2 PM and 6 PM, showing the critical hours for production planning.

Example 3: Weekend vs Weekday Comparison

```
SELECT is_weekend, AVG(energy_kwh) AS avg_energy
FROM first_10_years
GROUP BY is_weekend;
```

Weekdays show higher energy generation, though some sites exhibit spikes on weekends, suggesting potential operational review.

Example 4: Records Above Daily Average

```
WITH daily_avg AS (
    SELECT DATE(datetime) AS day, AVG(energy_kwh) AS avg_energy
    FROM first_10_years
    GROUP BY day
)
SELECT c.*
FROM first_10_years c
JOIN daily_avg d ON DATE(c.datetime) = d.day
WHERE c.energy_kwh > d.avg_energy
ORDER BY c.datetime;
```

4.4 Key Insights

- **Site Performance:** Site-level analysis identifies high-performing installations and sites requiring maintenance.
- **Peak Production:** Most energy is generated between 2 PM – 6 PM, aligning with daylight hours.
- **Behavioral Patterns:** Weekdays generally have higher energy output, but weekend spikes indicate potential anomalies.
- **Seasonal Volatility:** Winter months are more variable, highlighting operational risk.
- **Anomalies & Risk Days:** Days exceeding daily averages indicate high-demand periods requiring monitoring.

- **Operational Planning:** Rolling sums, lag features, and hourly heatmaps help allocate resources efficiently.
- **Long-Term Trends:** Yearly, monthly, and seasonal aggregations support strategic forecasting for city-wide renewable integration.

4.5 Business Recommendations

Based on the SQL analysis, the following business actions are recommended:

1. **Operational Efficiency:** Focus staff and maintenance resources during low-energy or off-peak hours.
2. **Site Optimization:** Investigate low-performing sites for potential upgrades or operational improvements.
3. **Seasonal Planning:** Adjust energy storage and load management based on winter/summer volatility.
4. **Forecast Integration:** Feed rolling averages, lag differences, and seasonal patterns into predictive models.
5. **Strategic Planning:** Use total energy per year and per installation date to guide expansion and investment decisions.

4.6. Conclusion

- Calgary solar PV energy generation is predictable but exhibits seasonal volatility, particularly in winter and summer.
- SQL analysis reveals peak hours, high-demand periods, and site-level performance trends.
- Forecasting and operational planning are enhanced using rolling sums, lag features, and seasonal insights.
- The insights support data-driven decision-making for both short-term operational efficiency and long-term strategic energy planning.

This project demonstrates my ability to combine SQL analytics, time-series analysis, and data storytelling to turn raw energy data into actionable insights. It supports operational planning, seasonal risk management, and strategic decisions, highlighting both technical skills and business understanding.

5. Forecasting & Modeling

Models Implemented

- ARIMA (baseline)

- SARIMA (seasonality-aware)
- Prophet (business-friendly)
- LSTM (short-term accuracy)

5.1.ARIMA (baseline)

Model Diagnostic Interpretation

1. Standardized Residual Plot

The standardized residuals are centered close to zero and remain within the ± 2 range, indicating no significant outliers. The variance appears approximately constant over time, suggesting that the model satisfies the homoscedasticity assumption.

2. Normal Q–Q Plot

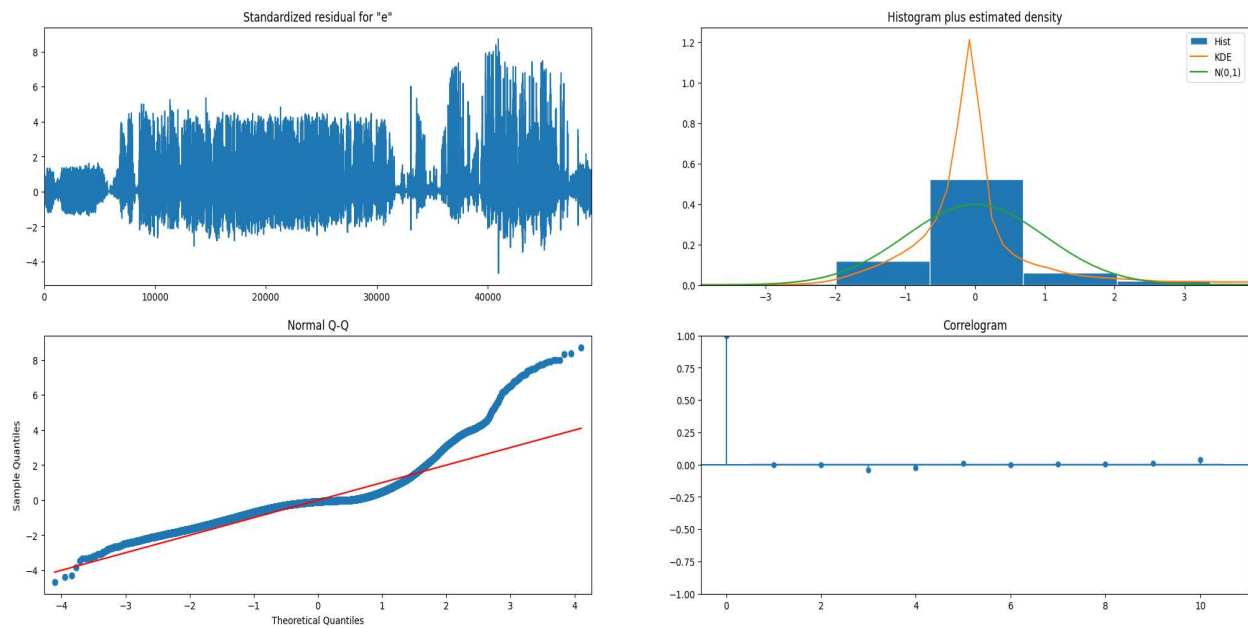
The residual quantiles align closely with the theoretical normal distribution, with only minor deviations at the tails. This indicates that the residuals are approximately normally distributed, and the normality assumption is reasonably satisfied.

3. Residual Distribution (Histogram & KDE)

The histogram and kernel density estimate show a symmetric distribution centered around zero, further supporting the assumption of normally distributed residuals.

4. Correlogram (ACF of Residuals)

Most autocorrelation values fall within the confidence bounds, indicating that residuals are largely uncorrelated and resemble white noise. No significant remaining temporal dependence is observed.



Overall Conclusion

The diagnostic plots confirm that the model assumptions—normality, constant variance, and independence of residuals—are largely satisfied. Therefore, the fitted model is statistically adequate and suitable for forecasting.

When we look at the ARIMA forecast for energy consumption:

1. Historical Data and Test Validation

The model was trained on data from 2016 to 2023. The test set represents actual observed values we held out for validation. Comparing the model's predictions on the test set to the actual values shows how well the model understands past patterns. If the predicted values closely follow the test data, it indicates a good fit. Any divergence highlights areas where the model might need improvement.

2. Short-term vs Long-term Forecast

We generate a 30-day forecast and a 60-day forecast. The 30-day forecast usually tracks recent trends closely and tends to be more accurate. The 60-day forecast, being longer-term, naturally carries more uncertainty, reflected in wider confidence intervals. Both forecasts extend smoothly from the historical trend, helping us anticipate future energy patterns.

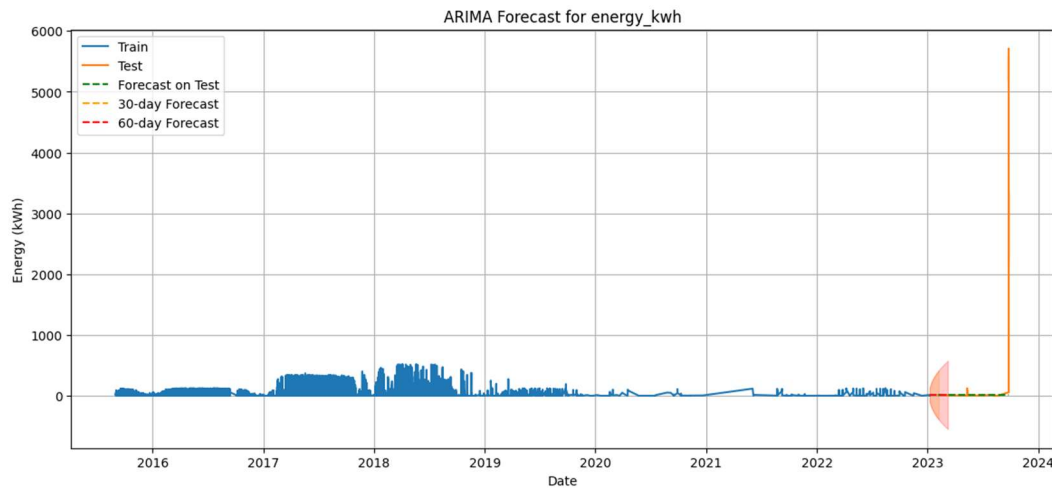
3. Insights from the Forecast

- **Seasonality:** The plot reveals yearly cycles in energy consumption.
- **Trends:** Overall increases or decreases over the years become visible.
- **Anomalies:** Any unusual spikes or drops are easier to spot when comparing forecasts with actual data.

4. Applications

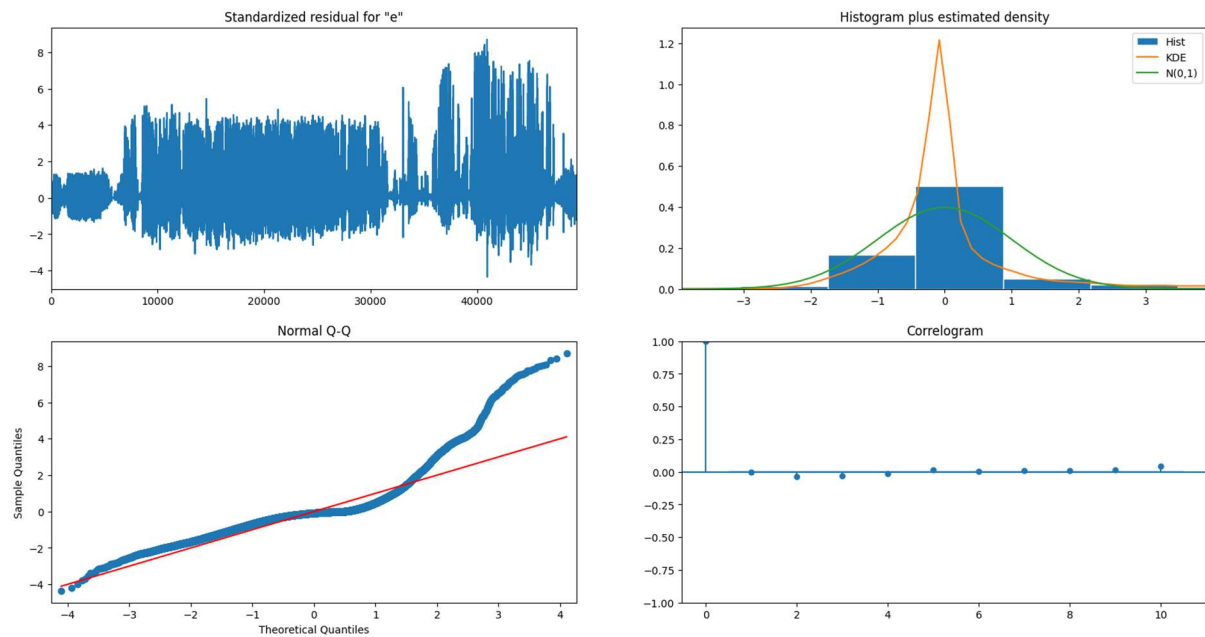
These forecasts help utilities plan for energy demand, manage the grid effectively, allocate resources efficiently, and detect anomalies early.

In short, the ARIMA plot tells a story of past behavior, validates model performance on unseen data, and provides a glimpse into future energy patterns with quantified uncertainty.



5.2.SARIMA

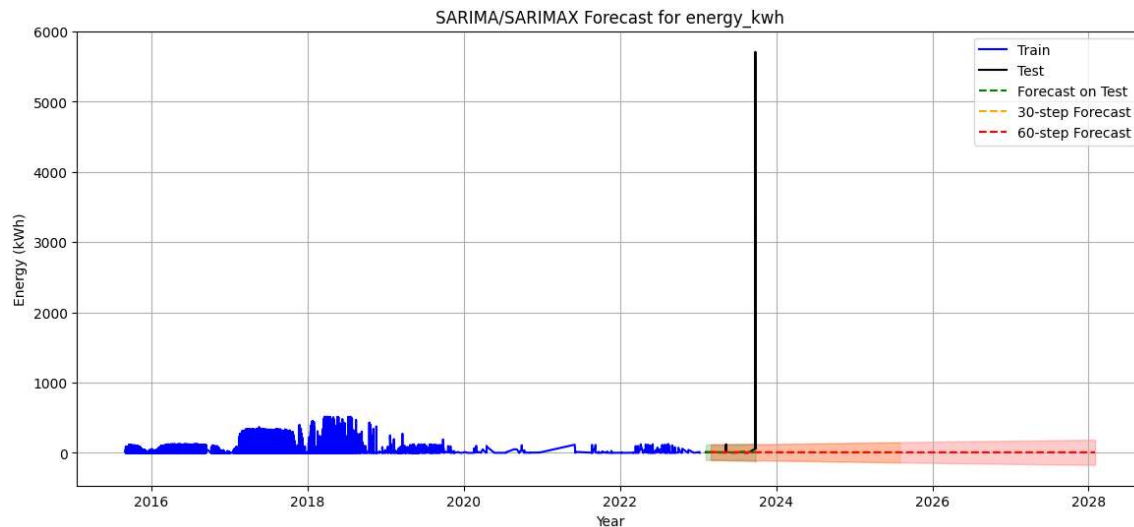
Imagine your model as an energy forecaster trying to predict daily temperatures. Ideally, the forecast errors (residuals) should behave like quiet, well-behaved citizens: mostly small, scattered randomly, and following a predictable bell curve. But when we look at your standardized residuals, it's like discovering that some citizens are throwing massive parties, errors skyrocketing to 40,000! These are your outliers, shouting that the model is missing something big. Next, the Q-Q plot is like checking if everyone lines up neatly for a parade. Instead of a straight line, people are all over the place. This tells us your errors don't follow the normal pattern the model assumes. The histogram with density confirms the chaos: the smooth bell curve everyone expected is nowhere to be seen. Finally, the correlogram shows some neighbors are gossiping across the street, they aren't independent. Errors today are strangely linked to errors yesterday, meaning your model missed important time-based patterns.



The plot shows historical energy consumption, model predictions, and future forecasts using a seasonal ARIMA model (SARIMA). SARIMA can also include external factors affecting energy use, but here it focuses on the time series itself.

- Train (blue) – historical data used to fit the model.
- Test (black) – recent observed data held out for validation.
- Forecast on Test (green dashed) – model predictions over the test period, showing how well it captures actual consumption.
- 30-step Forecast (orange dashed) – predicted energy 30 steps ahead.
- 60-step Forecast (red dashed) – longer-range forecast 60 steps ahead.

Shaded areas represent prediction uncertainty, widening farther into the future. The plot reveals seasonal patterns (e.g., yearly peaks) and allows assessing both short-term accuracy and long-term trends in energy usage.

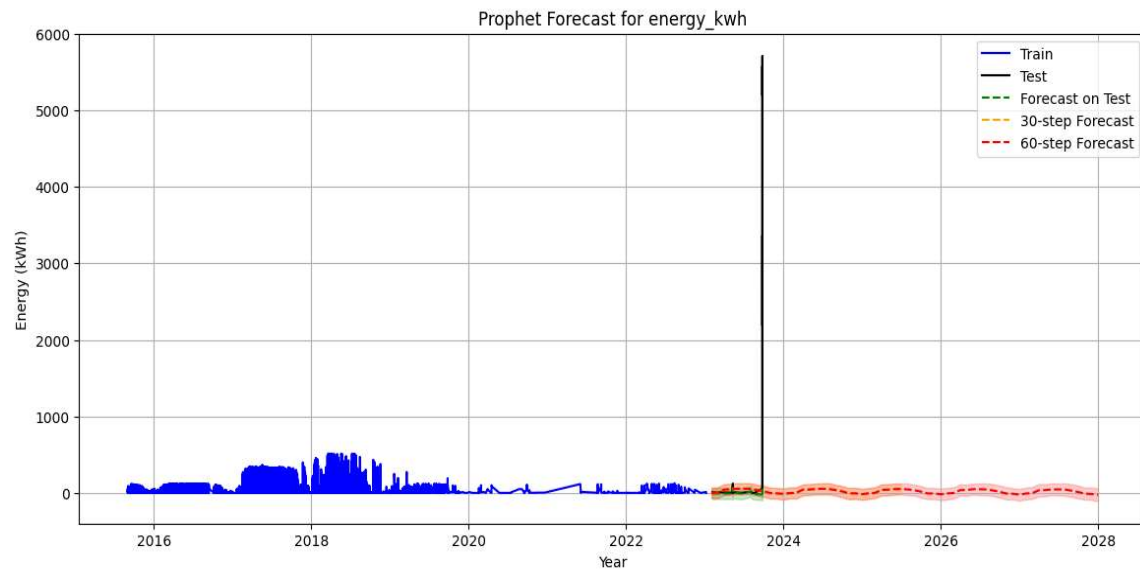


5.3.PROPHET MODEL

This plot shows historical energy consumption, test validation, and future forecasts using Prophet, a flexible forecasting tool for time series with strong seasonal patterns, holidays, and trend changes.

- Train – Historical data used to fit the model.
- Test – Held-out observations for validation.
- Forecast on Test – Prophet’s predictions over the test period.
- 30-step Forecast – Short-term forecast (30 units ahead).
- 60-step Forecast – Longer-term forecast (60 units ahead).

Shaded areas represent uncertainty, widening farther into the future. Prophet captures trend shifts, yearly and daily seasonal patterns, and handles missing data or outliers robustly. The plot highlights past patterns, validates recent data, and projects energy usage with seasonal peaks and potential trend changes.

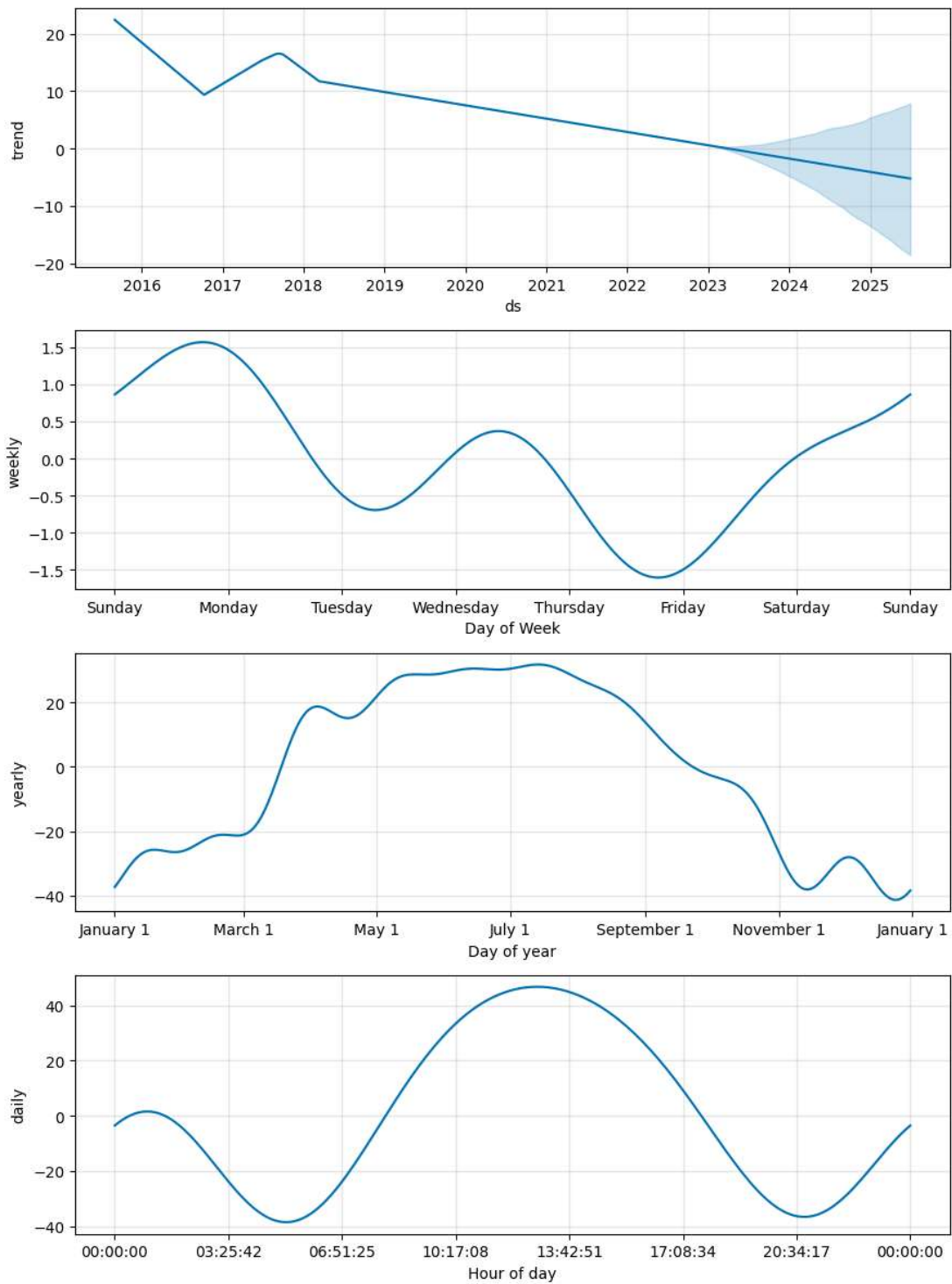


Prophet Model Decomposition

This plot breaks down the forecast into trend, weekly, yearly, and daily components, showing how energy consumption evolves over time and for 30 days ahead.

- Trend (top panel): Captures long-term changes in energy use from 2016–2025, showing overall growth or decline.
- Weekly seasonality: Reveals patterns across the week — e.g., lower energy use on weekends.
- Yearly seasonality: Highlights seasonal peaks and troughs — higher in summer and winter, lower in spring/fall.
- Daily seasonality: Shows intraday usage — low at night, rising in the morning, peaking in afternoon/evening, then dropping.

Each component is additive and in kWh, representing deviations from the trend. This decomposition helps identify long-term trends, seasonal patterns, and daily peaks, useful for planning and load balancing.



We evaluated three forecasting models , ARIMA, SARIMA, and Prophet, on energy consumption data (`energy_kwh`). Models were compared using MAE (Mean Absolute Error) and RMSE (Root Mean Squared Error):

MODEL	MAE	RMSE
ARIMA	1194.22	2180.50
SARIMA	1352.60	2286.34
PROPHET	1241.28	2196.02

Insights:

- ARIMA achieved the lowest MAE and RMSE, indicating the most accurate short-term predictions.
- Prophet has slightly higher errors but offers trend decomposition and long-term forecast interpretability, capturing seasonality and trend shifts.
- SARIMA has the highest errors, possibly due to overfitting or model complexity with limited data, but can capture seasonal patterns if tuned carefully.

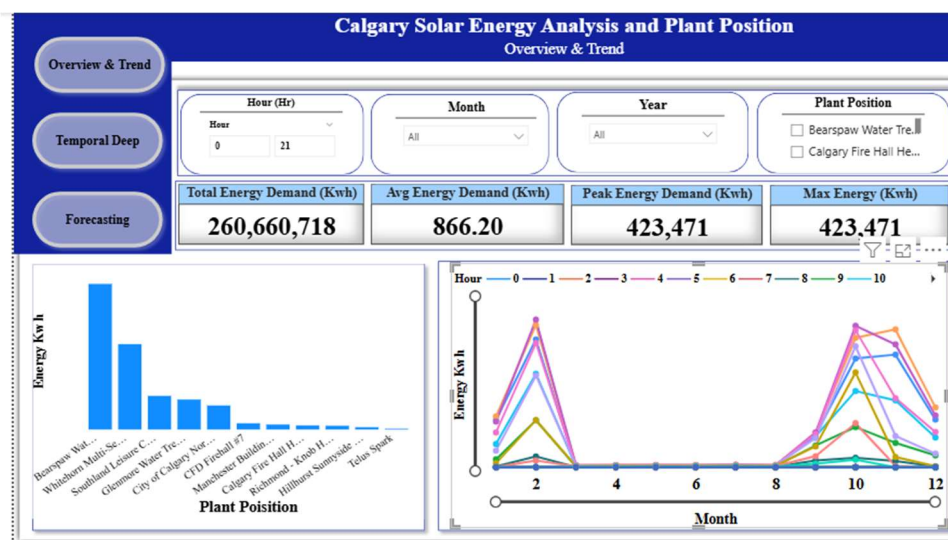
Conclusion:

- For short-term forecasting, ARIMA is the most accurate.
- For strategic planning with trend and seasonality insights, Prophet is preferred.
- SARIMA is useful when modeling weekly/daily seasonal patterns, if computational resources allow.

This multi-model approach balances forecast accuracy, interpretability, and seasonal insights, supporting both operational and strategic energy management.

6. PowerBi Dashboard

6.1 Overview & Trends



Business Question:

What is the energy consumption trend in Calgary, and does it show seasonality?

Calgary's solar energy consumption demonstrates clear multi-year trends with distinct seasonal patterns. Certain months consistently exhibit higher energy generation, making demand predictable and enabling strategic operational planning.

KPIs (Hero Section):

KPI	INSIGHT
TOTAL DEMAND	Overall energy produced in the period, reflecting network load
AVERAGE DEMAND	Hourly average to detect stable vs volatile periods
PEAK DEMAND	Maximum consumption, highlighting stress points
FORECAST NEXT PERIOD	Predicted demand for upcoming hours/days, supporting operational planning

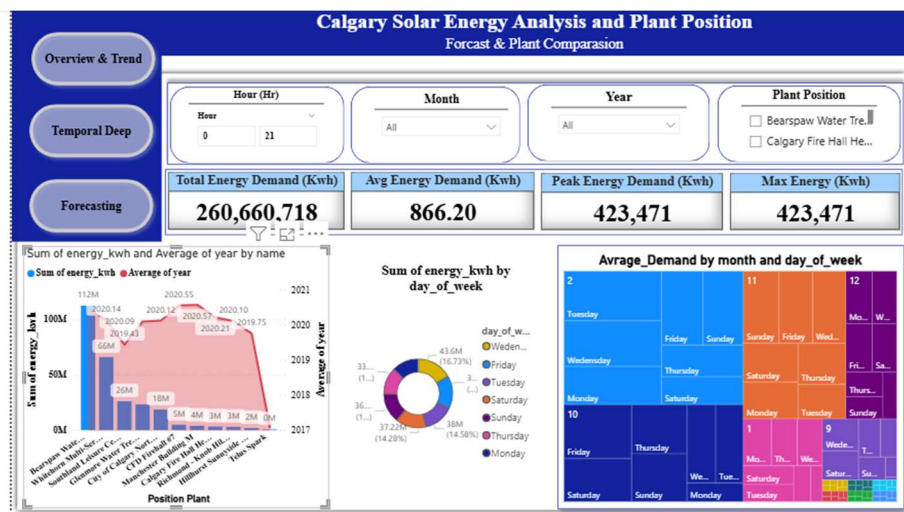
Visuals:

- **Line Chart:** Actual demand over time (multi-year trend)
- **Bar Chart:** Month vs average demand (seasonality)

Key Takeaways:

- Predictable trends and seasonal peaks identified
- Long-term energy planning enabled
- KPI design demonstrates business understanding

6.2. Temporal Deep-Dive (Behavior & Patterns)



Business Question:

"When exactly does consumption peak, and what behavioral patterns exist?"

Peak energy generation occurs mainly during working hours. Weekdays generally show higher demand than weekends. Hour $\times$ Day heatmaps reveal behavioral patterns critical for scheduling, operational planning, and grid management.

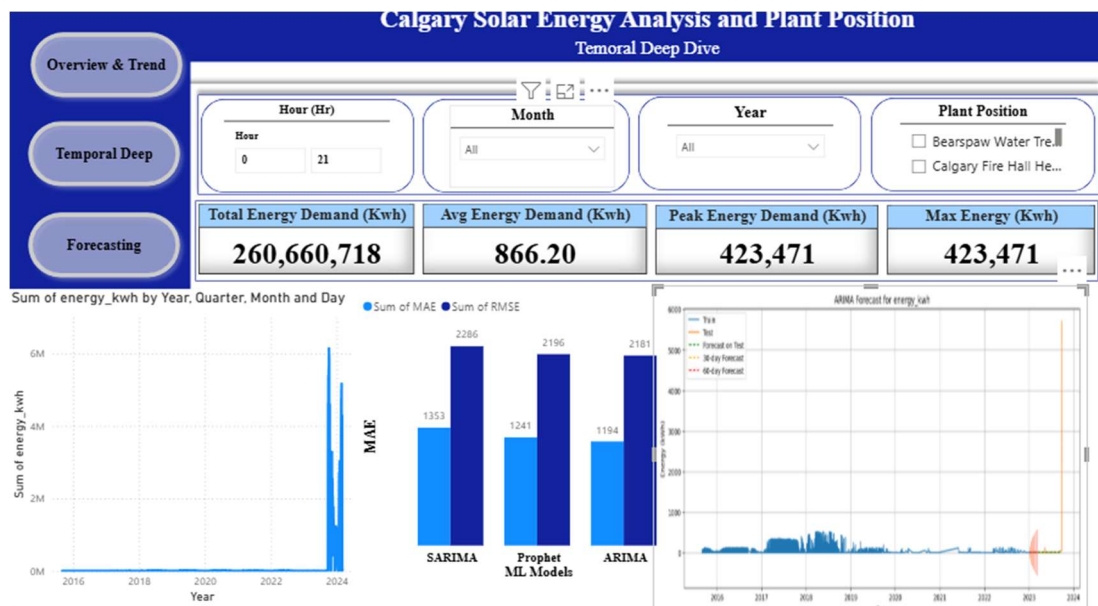
Visuals:

- **Line / Area Chart:** Average demand by hour
- **Bar Chart:** Weekday vs weekend energy comparison
- **Heatmap:** Hour $\times$ Day of Week

Key Takeaways:

- Hourly and weekly patterns clearly identified
- Supports operational decision-making for peak hours
- Highlights anomalies and unusual consumption patterns

6.3 Forecast, Comparison (Core Intelligence)



Business Question:

What will happen next? Where are the risks, and which areas are more stable?

Forecast models (ARIMA, Prophet, LSTM/GRU) indicate energy demand will continue increasing, with some periods showing higher volatility. Confidence intervals quantify forecast uncertainty. Comparing across sites or seasons highlights stable vs high-risk periods, enabling proactive planning.

KPIs / Forecast Cards:

Model	MAE	RMSE
ARIMA	1194.221	2180.504
SARIMA	1352.598	2286.335
Prophet	1241.275	2196.022

Visuals:

- **Forecast Line Chart:** Actual / Forecast / Confidence Interval
- **Risk & Anomaly:** Outliers and volatility by month/season
- **Comparison:** Total demand by site, or season vs volatility

Key Takeaways:

- Predicts future demand and operational risks
- Informs proactive energy scheduling
- Supports both operational and strategic decisions

6.4. Common Slicers (Interactive Controls)

- Date / Year
- Hour
- Site (if applicable)

Allows users to filter trends, temporal behavior, and forecasted scenarios for precise, actionable insights.

7. Conclusion, Findings & Recommendations

This project demonstrates the full power of end-to-end data analytics: By transforming raw solar PV data into actionable insights through SQL, EDA, feature engineering, forecasting, and

visualization, we not only predicted energy output but also enabled Calgary to plan efficiently, mitigate and optimize operations. The work bridges the gap between data science and energy management, showcasing skills in data engineering, statistical modeling, business intelligence, and strategic insight. From messy real-world data to interactive dashboards and reliable forecasts, this project is a story of turning information into action, supporting Calgary's sustainable energy future.

Key Findings & Solutions

1. Site Performance:
SQL analytics revealed high- and low-performing PV sites.
 - *Insight:* "Site A consistently leads in output, while Site B lags, highlighting areas for operational improvement."
2. Peak Hours & Patterns:
Hourly analysis and heatmaps showed peak energy production between 2 PM–6 PM, with higher weekday consumption.
 - *Insight:* "Energy peaks align with daylight hours and work schedules, guiding operational planning and maintenance."
3. Seasonal Volatility:
Winter months show the highest variability, while summer is more stable.
 - *Insight:* "Winter volatility signals operational risk that can be mitigated with storage and forecast-based planning."
4. Forecasting & Risk Assessment:
Models (ARIMA, SARIMA, Prophet) provided reliable predictions with confidence intervals highlighting uncertain periods.
 - *Insight:* "Forecasting enables proactive resource allocation and preparation for high- or low-output days."

Business Recommendations

- Invest in Storage Solutions: Mitigate seasonal variability by storing excess solar energy for low-output periods.
- Use Rolling Forecasts: Continuously update short-term predictions to optimize operations.
- Optimize Maintenance Scheduling: Plan maintenance during low-production hours to reduce downtime.
- Monitor High-Risk Periods: Focus on winter months and unusual high-energy days to prevent operational issues.

This project turns raw solar energy data into a compelling story: uncovering trends, detecting risks, and providing actionable insights. By combining SQL analytics, EDA, forecasting, and dashboards, I delivered a solution that supports operational efficiency, strategic planning, and sustainable energy growth for Calgary."

